

1. Hardware Layer (src-tauri/src/hardware/)

Mode S decoder: frames left-align into a u128 register so accessors use spec MSB-first numbering. CRC-24 with generator $x^{24} + \dots + x^3 + 1$; validity = remainder equals ICAO (address parity) or zero (even parity); single-bit repair scans 112 flips. Global CPR pairs even/odd fractions inside a 10 s epoch with the NL-table guard. Altitude: Q-bit arithmetic (x25 ft) or Gillham gray (x500 ft). Velocity ME TC19 subtypes 1/2 give vector ground speed and track; 3/4 give heading and airspeed; vertical rate is a shared 9-bit x64 ft/min field.

ACARS: SOH MODE TAIL ACK LABEL BID [STX FLIGHT CR TEXT] ETX BCS16 grammar with CRC-16/CCITT-FALSE (reference vector 123456789 -> 0x29B1). Weather classification routes METAR/SPECI prefixes and D-ATIS label shapes. `synthesize_frame()` is public so the simulator and fixtures build wire-exact frames.

Registry: three channels bind to factory serials via `ChannelPlan`; `UsbProbe` trait offers a `sysfs` libusb-free enumeration and a simulated inventory; mismatched hardware renders channels offline instead of refusing boot.

1.1 Simulator scenarios

- NAF911/912 intercept geometry solved analytically: crosses VL604 path at $T+70$ s, ~1 km miss (guaranteed STCA).
- VL604 squawk 7700 at $T+45$ s (event + TC-28 status frame); dropout windows exercise coasting.
- 0BADCO dark target transmits positions only; parity alternates per emitted frame (not per tick).
- `xorshift64*` seeded: same seed reproduces the byte-identical frame timeline.

2. Kinematics Layer (src-tauri/src/kinematics/)

`linalg.rs`: row-major fixed-size kernels, Cholesky factorisation returning `None` on non-positive pivots, multi-RHS solving used for both Kalman gains and Mahalanobis distances. `ekf.rs` (historic name, linear KF): 6-state Kalman filter (CV), $\sigma_a = 3.5 \text{ m/s}^2$ default, Joseph-form updates, gate $\chi^2(3)=16.27$ engaged after 12 accepted fixes. `dead_reckoning.rs`: `SiteOrigin` series-expanded metres-per-degree, leader-line projection, analytic closest approach. `rtree.rs`: full R^* (overlap-enlargement `ChooseSubtree`, margin-minimising split, 30% forced reinsertion), arena-backed. `stca_math.rs`: corridor bulk-load, candidate retrieval, 5 s time-stepped refinement requiring simultaneous infringement.

3. Weather Fusion (src-tauri/src/weather_fusion/)

BDS 4,4: WS[0..12] kt, WD[12..23] x360/2048 deg, validity flag, SAT[24..40] x0.25 C. BDS 4,5: turbulence/shear/microburst pairs, icing state with all-invalid sentinel, ISA deviation x0.1 C. Encoders and decoders round-trip. `spatial_interp`: ISA atmosphere (lapse 6.5 C/km, isothermal above tropopause), IDW with altitude penalty, vector-averaged wind, ISA-blended pressure, confidence from diversity x range; dust layer from visibility plus thermal bump above 28 C. Fusion engine: 512-node deque, 15 min TTL, priority chain AWOS > D-ATIS > METAR (first writer wins a gap, AWOS always overwrites).

4. Defence (src-tauri/src/defense/)

AnomalyDetector: emergency squawk mapping (7700/7600/7500), identity-less emitter flagging after 30 s arming, absentee silence after 45 s; transition-only verdicts; squawk churn counted but never alerting alone. GeofenceMonitor: ray-cast polygons with vertical bands, entry-only events with occupancy sets, runtime activation toggles. Intercept: damped fixed-point pursuit iteration (tolerance 15 m, ≤ 48 iterations) with forward-simulation feasibility check; headings measured from north matching engine convention (a from-east/from-north mix once flew the interceptor north when ordered east - cardinal cases now pinned by tests).

5. Engine & Host (engine.rs · sidecar.rs · commands/)

SurveillanceEngine: queued ingestion drained after the single per-tick prediction pass (the queue-then-drain structure exists because fusing inside ingest double-propagated filters and locked the acquisition gate out of reality); periodic scans STCA 15 Hz, anomaly 1 Hz, geofence 2 Hz; alert composition folds squawk family > dark verdicts > geofence hold-down > manual override; snapshots publish on a tokio broadcast bus as Arc<EngineSnapshot>.

Sidecar bridge: NDJSON stdio (init/log_tracks/shutdown; ready/awos/db_ack), writer task serialising child stdin, graceful None on spawn failure. Commands: twelve synchronous Tauri handlers over Arc<Mutex<Engine>> with sub-millisecond lock discipline; compute_intercept converts the frontend km/h to solver knots at the boundary.

6. Frontend (src/)

Telemetry store: 60 Hz snapshots land in a module singleton; the radar canvas reads the ref inside its own rAF loop (zero React churn), panels sample at 2-8 Hz. SI policy enforced through lib/units.ts (km rings 100/200/300, metres altitudes, km/h speeds, Doc-4444 minima rendered 9.26 km / 305 m). Browser demo feed mirrors the Rust roster at 20 Hz for offline operation and screen capture.

6.1 Wire contracts (models.rs mirror)

```
Track { icao24, callsign, class, alert, latitude, longitude,
        altitude_ft, ground_speed_kt, course_deg, vertical_rate_fpm,
        vertical_trend, squawk, on_ground, last_update_ms, age_s,
        coasting, position_sigma_m, leader_line[[lat,lon];13] }
EngineSnapshot { tracks, flight_data_blocks, stca_alerts,
                  geofence_breaches, anomalies, weather, status }
```

Wire values stay in DO-260B native units (feet/knots); conversion to SI happens only in presentation code. Field names are serde snake_case exactly.

7. Test Inventory

Suite	Count	Highlights
mode_s_decoder	12	CRC reference frames, repair, CPR round-trips, Gillham
acars_decoder	5	framing, checksum corruption, classification
sdr_registry	3	serial binding, offline degradation
simulator	10	100% decode rate, truth coordinates, determinism, STCA geometry
kinematics	17	Kalman filter convergence/gating, R*-tree integrity, STCA cases
weather_fusion	14	ISA values, IDW, priority chain, BDS round-trips
defense	16	squawk mapping, geofence cycles, intercept envelope
engine	10	110 s end-to-end: identity, STCA, coasting, breaches
sidecar bridge	6	codec round-trips, spawn degradation

python sidecar	21	AWOS codec, METAR, DuckDB durability
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